Course Name: Computer Networks and Internet Protocol

Assignment - Week 8 (Jan 2026)

TYPE OF QUESTION: MCQ/MSQ

Number of questions: 10 **Total mark: 10 X 1 = 10**

QUESTION 1

How does the router decide the destination if multiple entries in routing tables match the destination address?

a) It picks the first one
b) It picks the destination with the longest prefix matching
c) It picks the destination with the shortest prefix matching
d) It picks the last matched destination

Correct Answer: (b)

QUESTION 2

Which routing protocol is used for intra-domain routing in the Internet?

a) Border Gateway Protocol (BGP)
b) Routing Information Protocol (RIP)
c) Open Shortest Path First (OSPF)
d) Enhanced Interior Gateway Routing Protocol (EIGRP)

Correct Answer: (c)

Detailed Solution: Open Shortest Path First (OSPF) is a routing protocol used for intra-domain routing in the Internet. It is a link-state routing protocol that calculates the shortest path between routers based on the network topology.

QUESTION 3

In the BGP protocol, UPDATE and NOTIFICATION messages are used for:

a) Exchanging reachability information and to notify an error.
b) Exchanging reachability information and Confirming a BGP connection
c) Ensuring that a BGP neighbor is still alive and Confirming a BGP connection

d) Opening a BGP connection and Closing a BGP connection

Correct Answer: (a)

QUESTION 4

Why do we consider dividing an IP address into network address and host address?

a) To increase the total number of IP addresses possible.
b) Routers route the packets based on the host address.
c) To avoid the overhead of storing all possible host IP addresses in each router.
d) For resolving IP addresses from the domain name.

Correct Answer: (c)

Detailed Solution: By using the network address, routers can forward packets like the postal system (country, state, city,..), by looking at the relevant network address only instead of the entire host address. Thus routers do not need to store all possible host IP addresses.

QUESTION 5

Consider a network with three routers, A, B, and C. The routing table for router A has the following entries:

Destination: 192.168.1.0/24, Next Hop: B

Destination: 192.168.1.128/28, Next Hop : D

Destination: 192.168.2.0/24, Next Hop: C

If router A receives an IP packet with a destination address of 192.168.1.10, what will be the next hop for this packet?

a) D
b) B
c) C
d) Broadcast to all routers

Correct Answer: (b)

Detailed Solution: Since the destination address of the IP packet is in the 192.168.1.0/24 subnet, router A will forward the packet to the next hop router B.

QUESTION 6

When you connect your personal computer (end device/host) to the internet, how does it know its own IP address, and the Gateway IP address to use for sending packets to remote hosts, without any manual configuration?

a) Domain Name System (DNS)
b) From its own routing table.
c) Address Resolution Protocol (ARP)
d) Dynamic Host Configuration Protocol (DHCP)

Correct Answer : (d)

QUESTION 7

Which of the following data structures are used for speeding up the Longest Prefix Matching based Forwarding algorithms in routers?

a) Double Linked List
b) Segment Trees
c) Red Black Trees
d) Patricia Trees

Correct Answer: (d)
Detailed Solution: Patricia Trees are useful for computing the longest prefix search efficiently by storing the routes bit by bit starting from the root to the leaves. Different types of traffic to ensure that they are delivered fairly based on their importance.

QUESTION 8

Which is TRUE about BGP routing protocol?

a) It is an Intradomain routing protocol.
b) It is a type of Link State routing protocol
c) BGP relies on IGP for packet forwarding between IBGP peers.
d) BGP replaces the IGP protocol in an AS.

Correct Answer: (c)

Detailed Solution: Since the rate of token arrival is the same as the rate of packet arrival, the bucket will always have enough token to send the packets at the same rate they arrive. Therefore there will not be any burst.

QUESTION 9

Suppose we have *n* number of routes in a routing table. A Packet bearing a certain IP arrives at the router and requires it to be routed to the appropriate next hop location. Assuming that the router maintains a Patricia Tree for the routing table and we are using IPV4 addresses, what is the time complexity of this forwarding?

a) $O(n^2)$
b) O(32)
c) O(n)
d) O(logn)

Correct Answer: (b)

Detailed Solution: For finding the Longest Prefix Match we need to travel from the root of the Patricia Tree down to the node bearing the longest matching prefix. Since for this we need to travel at most the height of the tree. This is the utmost number of bits in address i.e. 32. So O(32).

QUESTION 10

Consider in distance vector routing protocol, router A has the routing table (only the relevant portion of the table is shown):

To	Cost	Next Hop
Z	8	B
….	….	….

After this, A receives the routing table information from C with the following:

To	Cost
Z	6
….	….

After this, A also receives the routing table information from D with the following:

To	Cost
Z	8
….	….

What will be Cost and Next Hop for Z in the routing table of A after this?

Consider the cost of each link is 1. (cost = number of hops required)

a) 8, B
b) 6, C
c) 9, D
d) 7, C

Correct Answer: (d)

Detailed Solution: The lowest cost to Z is coming from C which is 6. So,

Cost(Z from A) = Cost(Z from C) + Cost(C from A) = 6 + 1 = 7.